

RAHIL SHARMA

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EDUCATION

Vrije Universiteit Amsterdam (VU), Amsterdam, NL (Dual Enrollment)

Master of Science (M.Sc.) in Computer Science (Big Data Engineering)

Sept 2024 – Jul 2026

Coursework: Information Theoretic Learning, Parallel Programming for Large Scale Systems, Software Asset Management, Security and Machine Learning, Web Data Processing Systems, Data Mining Techniques

GPA: 3.6/4

University of Amsterdam (UvA), Amsterdam, NL (Dual Enrollment)

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GPA: 3.6/4

International Institute of Information Technology, Bangalore, Karnataka, IN

Executive PG Program in Software Development with Specialization in Full Stack Development

Aug 2022 – Oct 2023

Coursework: Object Oriented Programming, Data Structures and Algorithms, User Interface and Front-End Development, Backend Development, Software Deployment, System Analysis and Design

GPA: 3.3/4

Symbiosis Institute of Technology, Symbiosis International University, Pune, Maharashtra, IN

Bachelor of Technology (B. Tech), Electronics and Telecommunication Engineering

Aug 2018 – Jun 2022

Coursework: Control Systems, Digital Electronics, Artificial Intelligence, Machine Learning, Communication Networks, Data Science, Deep Learning, Computer Architecture, Wireless Communication, Antennas and Wave Propagation

Percentage: 74%

WORK EXPERIENCE

Centrum Wiskunde & Informatica (CWI), Amsterdam, NL

Research Assistant: CSY Group (Security and Machine Learning)

Feb 2025 – Present

- Implemented protocols for Bucket Oblivious Sort (BOS), Oblivious Random Permutation (OSORT), and Oblivious Bloom Filter Insertion (OBFi) in Python with gRPC, aligned closely with formal pseudocode.
- Addressed challenges in dummy element management, client memory bounds ($\leq 2Z$), and encrypted data routing, ensuring theoretical guarantees held in practice.
- Debugged and validated critical phases (e.g., butterfly routing, collaborative mergesort, batch-based Bloom filter insertion) through reproducible logs and parameter sweeps.
- Currently extending the project into a sensitivity study of performance under varying bucket sizes, padding strategies, and false-positive bounds, bridging theoretical analysis with experimental evaluation.
- Authored multiple pseudocode versions and test harnesses clarifying client vs. server responsibilities, paving the way toward a first-author publication as well as exploring the protocol from an applications angle.

Vrije Universiteit, Amsterdam, NL

Senior Teaching Assistant

Dec 2024 – Present

- Courses: Learning Machines (Robotics), Data Mining Techniques, Information Theoretic Learning, Software Asset Management, Security & Machine Learning, Fairness in Data Science, Project Collective Intelligence, Basics of Programming, Project Development Workshop, Arduino/Tinker Lab Workshop (University of Twente).
- Supporting students in understanding core concepts, designing experiments, and analyzing results.
- Facilitating group projects and lab sessions to enhance collaborative and hands-on learning.
- Conducting one-on-one sessions to mentor students struggling with core concepts and project deliverables.
- Assisting in the creation of innovative teaching materials and evaluation rubrics to ensure fair and comprehensive assessment.

Precisely – US Based MNC, Mumbai, IN

Software Support Engineer I: Data Integration

Jan 2022 – Aug 2024

- Specialized in **Big Data** with a focus on seamless upgrades and personalized user support, optimizing customer experiences across Public Cloud platforms (Azure, AWS, GCP) and providing robust IaaS client support.
- Conducted thorough **Root Cause Analysis** to identify and resolve complex data issues, ensuring minimal downtime and maintaining system integrity.

- Led efforts to resolve critical production incidents, achieving leading **Mean Time to Resolution (MTTR)** and ensuring the highest levels of service continuity.
- Utilized advanced **data analysis** techniques to generate actionable insights, driving continuous improvement and enhancing system performance.
- Managed **cluster setups** and **HDFS** configurations, ensuring efficient data storage, processing, and accessibility across distributed systems.
- Extensive experience with a wide range of technologies, including database management software (MS-SQL Server, Oracle) and Mainframe technology software (e.g., IBM AS-400), coupled with continuous training to address evolving client requirements effectively.
- Focused on **personal and team upskilling** by conducting knowledge transfer sessions, ensuring the team stays updated with the latest technologies and best practices.
- Consistently delivered top-tier **Customer Satisfaction (CSAT)**, maintaining excellent communication channels and securing high CSAT scores by promptly addressing client issues.
- Received multiple accolades such as **Spot Awards, Stellar Performer Award, Round of Applause Award, and Joy Giver Award**, reflecting exceptional performance and contributions.
- Played a pivotal role in fostering a **collaborative** team environment, mentoring junior engineers, and contributing to team success through effective leadership and knowledge sharing.

TECHNOLOGICAL SKILLS

Programming & Scripting: Python, Java, JavaScript (Node.js), MATLAB, SQL, R, Bash, Shell

Data Science & Machine Learning: NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Keras, NLTK, OpenCV, Matplotlib, XGBoost, LightGBM

Big Data & Cloud Platforms: Snowflake, Databricks, Apache Kafka, AWS (Lambda, EC2, S3), Microsoft Azure, Google Cloud Platform (GCP)

Secure & Distributed Systems: Docker, gRPC, Cryptography Libraries (PyCryptodome), Homomorphic Encryption Concepts, Oblivious Data Structures (BOS/OBD/OBFI), Socket Programming, Parallel & Distributed Systems

Software Engineering & DevOps: Git, GitHub, GitLab, Bitbucket, REST APIs, CI/CD pipelines, Maven, Docker Compose

Web & Interface Development: Django, Flask, Bootstrap, Material-UI, HTML/CSS, RESTful API Design

Research & Experimentation Tools: EnergiBridge, RAPL Power Measurement, Statistical Analysis (ANOVA, Tukey HSD, Regression), Experimental Design (RCBD, Factorial), LaTeX, Overleaf, Jupyter Notebooks

Methodologies: Agile, Waterfall, CRISP-DM, Experimental Reproducibility, FAIR Data Principles

Functional Competencies: Teaching & Mentorship, Technical Writing, Research Design, Information Management, Analytical Problem Solving, Communication, Organization Management

Relevant Statistics, Calculus, Probability and Database Management Knowledge

CAPSTONE PROJECTS

Bucket Oblivious Sort (BOS) & Oblivious Bloom Filter Insertion (OBFI)

CWI Amsterdam | 2025

- Designed **privacy-preserving sorting and indexing protocols** under strict client-memory assumptions using gRPC, Python, and encrypted client-server communication.
- Implemented **multi-phase distributed algorithms** (bucket distribution, butterfly merge, collaborative mergesort) aligned with pseudocode from cryptographic literature.
- Verified correctness, memory bounds, and element security across Phases 0–5, achieving reproducible test results.
- Extended the architecture to batch-based oblivious Bloom-filter insertion, ensuring mathematical correctness of false-positive rates while maintaining encryption secrecy.

Energy-Performance Trade-offs of Java Garbage Collection Strategies

VU Amsterdam | 2025

- Conducted a **324-run empirical study** comparing Serial, Parallel, and G1 GC under controlled workloads using ExperimentRunner + EnergiBridge with Intel RAPL power counters.
- Designed a **Randomized Complete Block Design (RCBD) experiment, automated data orchestration** via Python, and analyzed results with **ANOVA and Tukey HSD tests**.
- Delivered statistically validated findings on energy efficiency and latency trade-offs between **OpenJDK and Oracle JDK** implementations.

NLP-Based Entity Linking, Recognition, and Fact Checking System

VU Amsterdam | 2024

- **Designed and implemented NLP pipelines** using Python, NLTK, and Transformers for **text processing and entity extraction**.
- **Integrated APIs like Wikidata and DBpedia** for **entity disambiguation and linking to knowledge bases**.
- Developed and tested fact-checking algorithms leveraging Explainable AI techniques to improve model transparency.
- Part of academic research focusing on enhancing real-world applications of Natural Language Processing in misinformation detection and information retrieval.

PACSHAP-FeatureGraph: Privacy-Preserving Feature Selection Framework

UVA Amsterdam | 2025

- Developed a **privacy-preserving feature selection framework** aligning with **GDPR data-minimization principles**, building on and extending Goldsteen et al. (2022).
- Combined **Knowledge Graphs** and **SHAP values** for **context-aware, explainable feature attribution**, addressing interpretability gaps in traditional decision-tree-based methods. Aim was to automatically select only necessary features while maintaining model accuracy.
- Implemented **PAC-Privacy guarantees** and **autoencoder-based encoding** to achieve rigorous data protection without compromising predictive performance.
- Conducted **empirical evaluations** demonstrating improved **privacy-accuracy trade-offs** versus state-of-the-art baselines.

- Designed an **automated feature-selection pipeline** supporting both numerical and categorical data, mitigating **cross-model privacy leakage** and **adversarial inference attacks**.

Smartphone-Based Activity Classification System

VU Amsterdam | 2025

- Built an activity recognition system using smartphone and smartwatch sensor data (iPhone 12, Apple Watch Series 7) to classify five physical activities: sitting, walking, running, cycling, and dancing.
- Engineered time and frequency-domain features from accelerometer, gyroscope, and heart-rate signals using sliding-window FFT and statistical aggregation techniques.
- Applied **PCA and forward feature** selection to reduce 59 extracted features to 8 optimal predictors, balancing interpretability and performance.
- Compared Random Forest, SVM, KNN, and LSTM models for both **classification and heart-rate regression**, finding Random Forest ($R^2 = 0.71$) outperformed LSTM ($R^2 = 0.41$) on limited tabular data.
- Implemented a complete data-preprocessing pipeline including z-score outlier detection, linear interpolation for missing values, and correlation-based redundancy reduction.

Medicus: The Smart Case Assistant

SCAAI | 2023

- Developed a predictive chatbot assistant for hospitals automating decision support, recovery tracking, and patient-feedback workflows.
- Winner of the AGBI Health-Tech Grand Challenge, sponsored by NITI Aayog and Mehta Hospitals.

Early Research and Publications

Symbiosis | 2020-2022

- Applications of Disease Detection using Digital Image Processing Techniques — **Library Philosophy & Practice Journal (University of Nebraska-Lincoln)**.
- A Novel Approach towards Healthcare using Identity Access Management and Machine Learning — **IJRASET (Top 2 TechBharat Hackathon)**.

INVITED TALKS

- Guest Mentor, AI Workshop on Machine Learning and Data Science**
Organized by Creative Technology, University of Twente April 2025
- Facilitator, Poster Sessions & Project Showcases**
Organized at Vrije Universiteit Amsterdam, NL June 2025
- International Student Ambassador**
University of Amsterdam (UvA) Supported orientation and integration for new international students Aug 2025
- Entrepreneurial Mindset and the Economics of Development**
Organized by the Great Lakes Institute of Management, Chennai, IN July 2022
- Role of an Entrepreneur towards Innovation**
Atal Great Lakes Balachandran Incubator, Pune, IN Jan 2022

PROFESSIONAL CERTIFICATIONS AND ACHIEVEMENTS

- SnowPro Core (COF-CO2) — Snowflake
- Azure Fundamentals (AZ-900); Azure Data Engineer Associate — Microsoft
- AWS Solutions Architect Associate — Amazon Web Services
- Databricks Lakehouse Data Engineer Associate — Databricks
- Confluent Developer for Apache Kafka — Confluent
- Google Cloud AI Essentials — Google
- Salesforce AI Associate — Salesforce
- Winner, AGBI Health-Tech Grand Challenge — NITI Aayog & Atal Innovation Council (2022)
- Top-2 Finalist, TechBharat National Hackathon — BMS College of Engineering (2021)
- Top-5 Recognition, Youth Zest Virtual Contest — Indian Institute of Project Technology (IIPT) (2021)

LEADERSHIP & COMMUNITY ENGAGEMENT

- Languages: **Dutch (A2), French (A1 – DELF A1)**, English (Full professional proficiency)
- Bronze Medalist** in the International French Language Olympiad organized by the French Embassy of India.
- Selected to represent India and my College as a Delegate in the **Conference of Harvard Project for Asian and International Relations (HPAIR)**. Networked and took active participation in all keynote sessions surrounding Leadership, Healthcare, and Social Causes.
- Finalist in the ZS Campus Beats Case Challenge organized by ZS Associates**. Selected as the Top 100 teams selected among 4000+ teams participated from various colleges all over India.
- Founder, **ANR Medicus Technologies Pvt Ltd** — **Healthcare startup focused on accessible, data-driven medical solutions (2021–2022)**
- Recipient of the “Best Fresher Award” at Meridian School (2018), recognized for exceptional presentation, charisma, and campus presence.